

BAYANDELGER, Mongolia

WMO: 443520

Lat: 45.7303N

Lon: 112.3553E

Elev: 1101

StdP: 88.78

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.3	-26.4	-31.3	0.2	-27.9	-29.4	0.3	-26.0	13.2	-17.3	12.0	-16.5	3.4	320	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.6	19.8	30.5	18.9	28.6	17.8	21.7	29.9	20.3	28.0	19.1	26.6	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	16.0	26.1	17.7	14.5	24.6	16.5	13.4	23.0	69.4	30.1	64.0	28.3	59.5	26.5	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.7	9.8	-31.2	35.6	2.5	2.1	-33.0	37.1	-34.4	38.4	-35.8	39.6	-37.6	41.1		
(5)			-31.3	23.0	2.3	2.1	-32.9	24.5	-34.3	25.8	-35.6	26.9	-37.3	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-18.6	-13.9	-4.5	5.2	12.7	18.8	21.8	20.0	13.5	4.1	-7.8	-16.1	(6)
(7)		DBStd	15.00	4.65	5.64	6.27	5.35	5.14	4.12	3.34	3.59	4.73	5.36	6.43	4.61	(7)
(8)		HDD10.0	3757	887	668	451	160	32	1	0	0	21	193	536	809	(8)
(9)		HDD18.3	5856	1146	902	709	394	185	43	6	23	154	442	786	1067	(9)
(10)		CDD10.0	1210	0	0	1	16	115	264	367	311	126	10	0	0	(10)
(11)		CDD18.3	267	0	0	0	0	10	57	114	75	11	0	0	0	(11)
(12)		CDH23.3	2621	0	0	0	3	157	586	1092	684	97	2	0	0	(12)
(13)		CDH26.7	881	0	0	0	0	38	194	412	218	18	1	0	0	(13)
(14)	Wind	WSAvg	4.2	3.8	3.9	4.6	5.3	5.4	3.9	3.4	3.5	4.1	4.3	4.0	3.8	(14)
(15)	Precipitation	PrecAvg	182	1	1	3	6	18	40	51	32	16	7	4	2	(15)
(16)		PrecMax	377	2	4	10	34	72	106	161	110	61	28	10	6	(16)
(17)		PrecMin	91	0	0	0	0	1	8	9	1	1	0	0	0	(17)
(18)		PrecStd	69	1	1	2	8	20	23	37	26	15	7	3	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.8	2.9	14.8	23.1	30.4	34.1	36.0	34.5	28.8	20.9	9.2	-2.2	(19)
(20)			MCWB	-6.9	-0.8	6.1	12.1	15.1	20.1	21.0	22.4	16.5	12.0	3.8	-4.7	(20)
(21)		2%	DB	-7.9	-0.5	10.6	19.6	27.3	31.2	33.3	31.2	26.2	17.3	6.0	-5.6	(21)
(22)			MCWB	-9.3	-3.0	3.7	9.7	14.1	18.3	20.4	20.1	15.3	9.8	1.6	-7.2	(22)
(23)		5%	DB	-10.1	-3.2	7.3	17.0	24.7	29.0	30.9	29.0	23.7	14.5	3.3	-7.8	(23)
(24)			MCWB	-11.2	-5.3	1.9	8.2	13.1	17.4	19.5	18.6	14.4	7.9	-0.2	-9.1	(24)
(25)		10%	DB	-12.1	-5.7	4.7	14.4	22.2	26.9	28.8	27.1	21.4	12.1	0.8	-9.7	(25)
(26)			MCWB	-12.9	-7.3	0.0	6.7	11.8	16.4	18.7	17.5	13.1	6.2	-2.0	-10.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.7	-0.3	6.5	13.6	16.9	21.5	24.7	23.9	18.0	13.4	4.6	-4.5	(27)
(28)			MCDWB	-5.0	1.7	14.1	20.5	25.2	30.7	33.2	31.3	26.2	20.0	8.7	-2.4	(28)
(29)		2%	WB	-9.2	-3.0	4.0	10.6	15.3	19.9	22.1	21.5	16.4	10.2	1.9	-7.2	(29)
(30)			MCDWB	-7.9	-0.5	10.1	18.5	25.4	28.6	30.3	29.4	24.6	16.4	6.1	-5.8	(30)
(31)		5%	WB	-11.1	-5.1	2.1	8.6	13.9	18.6	20.8	19.9	15.1	8.3	0.1	-9.0	(31)
(32)			MCDWB	-10.1	-3.2	7.2	16.1	23.7	26.8	28.4	27.2	22.4	14.1	3.6	-7.9	(32)
(33)		10%	WB	-12.9	-7.3	0.3	6.8	12.4	17.3	19.5	18.3	13.8	6.5	-1.6	-10.6	(33)
(34)			MCDWB	-12.1	-5.7	4.6	14.3	21.2	24.8	26.8	25.4	20.4	11.8	1.1	-9.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	9.3	10.9	12.1	12.6	11.1	10.7	11.0	11.3	10.5	8.9	7.6	(35)
(36)			MCDBR	9.1	11.4	13.4	14.9	15.8	13.2	13.2	13.6	12.9	11.0	8.9	(36)	
(37)			MCWBR	8.3	9.4	9.2	9.4	8.6	6.8	6.1	6.8	7.7	8.5	8.2	7.8	(37)
(38)		5% WB	MCDBR	9.1	11.3	13.1	14.3	14.6	12.1	11.9	12.4	12.5	12.6	10.7	8.7	(38)
(39)			MCWBR	8.3	9.4	9.2	9.6	8.8	7.0	6.5	7.4	7.8	8.8	8.1	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.255	0.279	0.316	0.371	0.429	0.448	0.411	0.386	0.348	0.309	0.271	0.256	(40)	
(41)		taud	2.336	2.284	2.233	2.121	1.989	2.044	2.221	2.259	2.301	2.345	2.349	2.307	(41)	
(42)		Ebn at Noon	862	903	910	884	839	823	851	860	865	854	831	814	(42)	
(43)		Edh at Noon	74	96	118	146	172	164	136	126	110	90	73	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	3.22	4.73	5.98	6.69	6.63	6.43	5.91	4.85	3.46	2.25	1.68	(44)	
(45)		RadStd	0.17	0.20	0.19	0.26	0.34	0.40	0.25	0.25	0.22	0.19	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-1.38	N/A	-0.68	-0.83	N/A	N/A	N/A	N/A

Nomenclature: See separate page